ISHAN BHARDWAJ

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M.S. Financial Mathematics candidate. Top 10% finisher in the CME Institute Trading Competition (live IBKR execution, profitable all 5 OOS years). Built production-grade trading systems, options pricing engines, and credit risk models from scratch. Targeting systematic trading, quant research, and derivatives roles.

EDUCATION

University of Minnesota — Twin Cities M.S. Financial Mathematics — Expected Dec 2026

Stochastic calculus, derivatives pricing, financial econometrics, computational finance, statistical learning

KIIT University B.Tech, Computer Science — Jul 2020 – May 2024

SKILLS

Trading & Quant	Systematic strategy development, backtesting engine design, walk-forward IS/OOS validation, IBKR API (ib_insync), ES/NQ/CL futures, options pricing (Black-Scholes, binomial, Monte Carlo), Greeks, implied volatility, delta hedging
ML & Data	Python (pandas, numpy, scikit-learn, XGBoost), SHAP, pyarrow, asyncio, time-series forecasting, feature engineering
Finance	Cointegration analysis (Engle-Granger, Johansen), Kalman filtering, VaR/CVaR, Sortino/Calmar ratios, vol targeting, Kelly criterion

WORK EXPERIENCE

Quantitative Finance Expert (Contractor) — Jul 2026 – Present

Mercor

- Evaluate 40+ derivatives pricing, options Greeks, and structural credit model (Merton) problems weekly for a frontier AI research client — verifying Black-Scholes implementations, Monte Carlo convergence, and systematic trading logic against ground-truth solutions.
- Design benchmark problem sets for stochastic processes, risk model construction (VaR/CVaR), and portfolio optimization used directly in training and evaluation of frontier AI systems.

Equity Research Analyst Intern — Jun 2025 – Aug 2025

Atlasio Capital Partners

- Built DCF and scenario-based valuation models across 40+ companies; identified margin inefficiencies of 300–500 bps by decomposing revenue, cost structures, and leverage ratios at the segment level.
- Automated financial data ingestion and validation pipelines in Python (pandas + REST APIs), reducing manual data inconsistencies by 25% and improving research turnaround by 20%.

Quantitative Analyst Intern — Jan 2024 – Mar 2024

Mudraksh & McShaw

- Analyzed 500K+ transaction records to identify cost drivers and performance anomalies; built automated ETL pipelines reducing processing time by 20–25% and KPI dashboards tracking 15+ operational metrics.

Trade Support Analyst Intern — Nov 2023 – Dec 2023

AlgoBulls

- Validated 1,000+ daily transactions across algorithmic trading systems with near 100% accuracy; flagged pricing discrepancies and supported audit workflows across 10,000+ trades.

PROJECTS

CME Systematic Futures Trading System — Top 10%, CME Institute Competition — ISHAN217/CME-Futures-Trading-System

Python 3.12 · pandas · numpy · pyarrow · asyncio · IBKR API (ib_insync) · Custom backtesting engine

- End-to-end automated futures system across 5 independent strategies on ES/NQ/CL; IS/OOS ratio 1.34x, profitable in all 5 held-out years (2022–2026). Fill validity audit corrected win rate from 79.9% to honest 53–60%.
- Explored 15 signal types, rejected 11 after OOS failure with documented reasons; 38-check test suite verifying P&L to the cent.

Options Pricing Engine — From Scratch, No QuantLib — ISHAN217/options-pricing-engine

Python · Black-Scholes · CRR Binomial · Monte Carlo · Brent/Newton IV Solvers · Delta-Hedging Simulation

- Three independent pricers (BS analytical, CRR binomial w/ American early exercise, MC w/ antithetic variates for European/Asian/barrier) with all 5 Greeks verified to <1e-4 via finite-difference; IV round-trip <1e-6. Delta-hedging simulation validates vol carry formula.

Cointegration & Kalman Filter Pairs Trading — ISHAN217/stat-arb-pairs-trading

Python · Engle-Granger · Johansen Cointegration · Kalman Filter · Walk-Forward IS/OOS Backtest · Hurst Exponent

- 190 pairs screened across 5 sectors; Kalman filter for dynamic hedge ratio vs rolling OLS. All 8 IS-cointegrated pairs had negative OOS returns — documenting cointegration breakdown across 2021–2026 regime changes as the central research finding.